Multiplication: Rows, Arrays, and the Total
Answer Key
Grade 3

Quick Answers

1. 15	4. (b) = 32, —	7. (b) = 25, —
2. 24	5. (b) = 18, —	8. (b) = 56, —
3. 35	6. 4	

What is verified

4 of 8 problems fully machine-checked against SymPy.
— marks an answer only you can judge — problems 4, 5, 7, 8. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 3
3.OA.A.1 – *problems 1, 3, 4, 7*
3.OA.A.3 – *problems 2, 5, 8*
3.OA.A.4 – *problem 6*
Difficulty 1–4 of 5 across 12 tagged checks.

Common wrong answers

1. *If they got 8: added the rows and the dots in each row instead of multiplying*
2. *If they got 10: added the number of bags and the apples in each bag*
3. *If they got 5: counted only the rows and stopped*
4. *If they got 12: added the rows and the stickers in each row*
5. *If they got 9: added the boxes and the crayons in each box*
7. *If they got 10: added 5 + 5 instead of multiplying the 5 rows by 5 dots each*
8. *If they got 15: added the rows and the chairs in each row*

1. This array has 3 rows with 5 dots in each row. How many dots are there in all?
Solution: The total counts every dot, so multiply the rows by the number in each row: $3 \times 5 = 15$. Skip counting checks it: $5, 10, 15$. $\boxed{15}$

2. Nadia packs 4 bags with 6 apples in each bag. How many apples in all?

Solution: This is 4 equal groups of 6, so the total is $4 \times 6 = 24$. Adding the two numbers ($4 + 6 = 10$) would only mix the group count with the group size — it does not count the apples. $\boxed{24}$

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3. The chairs are in an array: 5 rows with 7 chairs in each row. How many chairs in all?

Solution: Each of the 5 rows holds 7 chairs, so the total is $5 \times 7 = 35$. Answering 5 would only name how many rows there are — the question asks for every chair. $\boxed{35}$

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4. Maya wrote $4 + 8 = 12$ for an array with 4 rows and 8 stickers in each row.

Solution (a): Maya added the number of rows to the number of stickers in each row. But the array shows 4 equal groups of 8 stickers, so the counts must be multiplied, not added — 12 does not count every sticker. A correct answer says the array means "4 groups of 8," so it needs 4×8.

Solution (b): The correct total is $4 \times 8 = 32$. $\boxed{32}$

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5. Leo wrote $6 + 3 = 9$ for 6 boxes with 3 crayons in each box.

Solution (a): Leo added the number of boxes to the crayons in each box. The story means 6 equal groups of 3 crayons, so the total needs multiplication — 9 counts a box number plus a crayon number, which is not a number of crayons.

Solution (b): The correct total is $6 \times 3 = 18$. $\boxed{18}$

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6. Each row holds 6 muffins and there are 24 muffins in all. How many rows? ($n \times 6 = 24$)

Solution: We know the total (24) and the number in each row (6); the unknown is the number of rows. Think of the related fact: $4 \times 6 = 24$, so $n = 4$. Check: 4 rows of 6 muffins is $4 \times 6 = 24$ muffins. $\boxed{n = 4}$

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7. Kai says "$5 + 5 = 10$" for an array of 5 rows with 5 dots in each row.

Solution (a): Kai's sum $5 + 5$ counts only two rows' worth of dots (or a row number plus a dot number) — it is not the total. The array has 5 rows and *each* row holds 5 dots, so all five rows must be counted.

Solution (b): The correct total is $5 \times 5 = 25$. Skip count to check: $5, 10, 15, 20, 25$. $\boxed{25}$

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8. Rosa wrote $7 + 8 = 15$ chairs for 7 rows with 8 chairs in each row.

Solution (a): Rosa's 15 counted the 7 rows and then the 8 chairs of a single row — a row count plus one row's chairs. That misses the chairs in the other six rows, so it cannot be the total.

Solution (b): Each of the 7 rows holds 8 chairs: $7 \times 8 = 56$. $\boxed{56}$